

1 DMC-MDE: Configuration in DMC

Overview

The MDE connection includes three basic components:

- The *MachineDataManager* component manages machine counters, actual cycles and sends data to the *HydraMDEWriter*. The component sends data at intervals or when statuses change. The *MachineStatusProvider* component informs about status changes.
- The *MachineStatusProvider* component manages machine statuses. There are different monitoring types inheriting from the *MachineStatusProvider* component ("no monitoring" type).
 - Cyclic monitoring (component: *CycleMonitoringMachineStatusProvider*)
 - Operating signal monitoring (component: *ProductionMonitoringMachineStatusProvider*)
- The *HydraMDEWriter* component writes dialog data for HYDRA to be processed in the server. Configure the *HydraMDEWriter* component as a child of the *MachineDataManager* component.

Use the *FileTransporter* component to transfer data files generated by the *HydraMDEWriter* to a shared directory on the server (transaction safety).

HydraMDEWriter

The *HydraMDEWriter* includes the following configuration parameters:

- **FileTimespan**
Specifies the frequency of generating a new data file for processing in the HYDRA server. Possible values:
 - Month: a new file once a month.
 - Day: a new file once a day.
 - Hour: a new file once an hour.
 - Minute: a new file once a minute.
- **OutputDirectory**
Specifies a local directory that stores the data files. If you indicate a relative path, the directory is identified based on the current runtime directory (%ProgramData%\mpdv\mdc\[DMC instance name]\...).

Standard *HydraMDEWriter* configuration:

```
<ComponentConfiguration Name="HydraMDEWriter"
  Class="mpdv.MachineDataCollector.Plugin.HydraPdv.HydraMDEFileWriter">
```

```

    <Parameter Name="FileTimespan" Value="Minute" />
    <Parameter Name="OutputDirectory" Value="MDE" />
    <Parameter Name="ExtensionTempFile" Value=".mda$" />
    <Parameter Name="ExtensionFinalFile" Value=".mdat" />-->
</ComponentConfiguration>

```

MachineDataManager: Counter configuration

The **MachineDataManager** component includes the following configuration parameters:

- **CounterConfiguration**

Refers to an XML file that includes the [Counter configuration](#).

- Specify the configuration as "[Type] + "://" + [relative path].
- The relative path is identified based on the current runtime directory (%ProgramData%\mpdv\mdc\[DMC instance name]_default\[Type]...).

SendingIntervalToHydra

Optional parameter; interval in seconds for the cyclic update of counter differences and actual cycles.

This parameter sends data to the configured *HydraMDEFileWriter* at the specified interval.

Default interval: 60 seconds

ConfigMDEJtp_ListServiceName

HYDRA *ListServiceName* to read out counter data from the HYDRA database. When starting the program, counter data is written into the above-mentioned XML file.

ConfigMDEJtp_JtpConnector

Name of the *JtpConnector* component you can use to connect to the HYDRA server. (By default: JtpConnector).

ConfigMDEJtp_RequestInterval

Interval in seconds to update counter data. You can only change existing counter data. (Default interval: 0 → no cyclic updating).

ConfigMDEJtp_ListServiceColumns

You can indicate the columns of the service. If no columns are specified, DMC requests the following columns from the HYDRA server.

- machine_counter.type
- machine_counter.monitoring
- machine_counter.designation"
- machine_counter.counter
- machine_counter.cyclecounter
- machine_counter.calculation_partitioning
- resource.id

These columns are currently required for correct functioning of the MDE. Standard *MachineDataManager* configuration:

```
<ComponentConfiguration Name="MachineDataManager"
  Class="mpdv.MachineDataCollector.Jis.MachineData.MachineDataManager"
  Children="HydraMDEFileWriter,OpcDriver">
  <Parameter Name="MachineDataManager_CounterConfiguration"
    Value="counter://counters.xml" /
  <Parameter Name="MachineDataManager_SendingIntervalToHydra" Value="20" />
  <Parameter Name="ConfigMDEJtp_JtpConnector" Value="JtpConnector" />
  <Parameter Name="ConfigMDEJtp_ListServiceName" Value="MDMachineCounter.list"
  />
  <Parameter Name="ConfigMDEJtp_RequestInterval" Value="5" />
</ComponentConfiguration>
```

The **counter configuration** includes the following configuration parameters:

Configuration of counters

Each CounterConfig section is a counter.

ID

Number of the counter (unique for each machine).

Text

Counter name.

CycleCounter

Stroke counters yes/no; HYDRA option: posting as cycles. If you use several stroke counters, the system adds up driver data.

Monitoring

Relevant to cyclic monitoring. Specifies if counters are applied for this purpose.

Type

Counter type

- Yield
- Scrap
- Rework
- Problem

Calculation_Partitioning

Currently not used.

```
<?xml version="1.0" encoding="utf-8"?>
<ArrayOfCounterConfig
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <CounterConfig>
    <Id>1</Id>
    <Text>Counter1</Text>
    <CycleCounter>true</CycleCounter>
    <Monitoring>true</Monitoring>
    <Type>Yield</Type>
    <Calculation_partitioning>1</Calculation_partitioning>
  </CounterConfig>
  <CounterConfig>
    <Id>2</Id>
    <Text>Counter2</Text>
    <CycleCounter>false</CycleCounter>
    <Monitoring>false</Monitoring>
    <Type>Scrap</Type>
    <Calculation_partitioning>1</Calculation_partitioning>
  </CounterConfig>
</ArrayOfCounterConfig>
```

This first DMC version does not access HYDRA's counter configuration. You have to configure counters twice, i.e. in the DMC and HYDRA.

MachineStatusProvider: status configuration and status monitoring

The **MachineStatusProvider** component includes the following configuration parameters:

- **MachineStatusConfiguration**
Refers to an XML file that includes the [status configuration](#).
- Specify the configuration as "[Type] + "://" + [relative path].
- The relative path is identified based on the current runtime directory
(%ProgramData%\mpdv\mdc\[DMC instance name]_default\[Type]...).

OutputMachineLock

Optional parameter; output for the machine lock to send a signal to the driver (machine). This configured output is set, if you defined the parameter "MachineLock" for the currently active status in the [status configuration](#).

OutputMachineDowntime

Optional parameter; output (machine down). This configured output is set, if the status class "production" is set for the currently active status.

ConfigMDEJtp_ListServiceName

HYDRA *ListServiceName* to read out status configuration from the HYDRA database. (By default: MDMachineCounter.List). Status configuration is written into the above-mentioned XML file.

ConfigMDEJtp_JtpConnector

Name of the *JtpConnector* component you can use to connect to the HYDRA server (by default: *JtpConnector*).

ConfigMDEJtp_RequestInterval

Interval in seconds to update status configuration. You can only change existing counter data (default interval: 0 → no cyclic updating).

ConfigMDEJtp_ListServiceColumns

You can indicate the columns of the service. If no columns are specified, DMC requests the following **columns** from the HYDRA server.

- statusassignment.machine_status
- statusassignment.set_automatically
- statusassignment.set_manually
- statusassignment.digitalinput_nr
- statusassignment.digitaloutput_machinelock
- statusassignment.productionlock
- statusassignment.production_characteristic
- statustext.text
- statusassignment.text_nr
- resource.id

These columns are currently required for correct functioning of the MDE.

Standard *MachineStatusProvider* configuration:

```
<ComponentConfiguration Name="MachineStatusProvider"
Class="mpdv.MachineDataCollector.Jis.MachineData.MachineStatusProvider"
Children="OpcDriver">
  <Parameter Name="MachineStatus_MachineStatusConfiguration"
Value="wpstatus://HeatingWorkplaceStatusConfiguration.xml" />
  <Parameter Name="MachineStatus_OutputMachineLock" Value="1" />
  <Parameter Name="MachineStatus_OutputMachineDowntime" Value="2" />
  <Parameter Name="ConfigMDEJtp_JtpConnector" Value="JtpConnector" />
  <Parameter Name="ConfigMDEJtp_ListServiceName"
Value="WorkplaceStatusAssignment.list" />
  <Parameter Name="ConfigMDEJtp_RequestInterval" Value="5" />
</ComponentConfiguration>
```

The **status configuration** includes the following configuration parameters:

Configuration of statuses

Each StatusConfig section is a machine status.

ID

Unique status number.

Text

Status name.

Class

Production identifier:

- N (used for "not assigned")
- P (stands for "production", defined once for each machine).
- S (stands for "malfunctions", n entries for each machine).
- A (stands for the "general disturbance", defined once for each machine).

Manual

Enable this option to enter statuses manually.

Automatic

Enable this option to enter statuses automatically.

ProductionLock

True enables the production lock (PSPERRE).

False disables the production lock (PSPERRE).

MachineLock

True enables the output "machine lock".

False disables the output "machine lock".

Input

The number of the digital input defines the input that identifies the status (0 = no input).

```
<?xml version="1.0" encoding="utf-16"?>
<ArrayOfStatusConfig
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <StatusConfig>
    <Id>1</Id>
    <Text>PRODUCTION</Text>
    <Class>P</Class>
    <Manual>false</Manual>
    <Automatic>true</Automatic>
    <ProductionLock>false</ProductionLock>
    <MachineLock>false</MachineLock>
    <Input>1</Input>
  </StatusConfig>
  <StatusConfig>
    <Id>2</Id>
    <Text>PREPARE</Text>
    <Class>S</Class>
    <Manual>true</Manual>
    <Automatic>true</Automatic>
    <ProductionLock>true</ProductionLock>
    <MachineLock>false</MachineLock>
    <Input>2</Input>
  </StatusConfig>
  <StatusConfig>
    <Id>99</Id>
    <Text>DISTURBANCE</Text>
    <Class>S</Class>
    <Manual>true</Manual>
    <Automatic>true</Automatic>
    <ProductionLock>true</ProductionLock>
    <MachineLock>true</MachineLock>
    <Input>3</Input>
  </StatusConfig>
  <StatusConfig>
    <Id>30000</Id>
    <Text>Not Assigned</Text>
    <Class>N</Class>
    <Manual>true</Manual>
    <Automatic>true</Automatic>
    <ProductionLock>false</ProductionLock>
    <MachineLock>false</MachineLock>
    <Input>0</Input>
  </StatusConfig>
</ArrayOfStatusConfig>
```

The **CycleMonitoringMachineStatusProvider** component provides the following configuration parameters in addition to the parameters of the parent component ([MachineStatusProvider](#)):

CycleTargetTime

Target cycle in seconds. The machine can overwrite the target cycle at runtime.

CycleMinimumTime

Minimum cycle time in seconds.

CycleExtension

Cycle extension in %.

Standard *CycleMonitoringMachineStatusProvider* configuration:

```
<!--<ComponentConfiguration Name="CycleMonitoringMachineStatusProvider"
      Class="mpdv.MachineDataCollector.Jis.MachineData.CycleMonitoringMachineStatusProvider"
      Children="OpcDriver">
  <Parameter Name="MachineStatus_MachineStatusConfiguration" Value="wpstatus://HeatingWorkplaceStatusConfiguration.xml" />
  <Parameter Name="MachineStatus_CycleTargetTime" Value="10" />
  <Parameter Name="MachineStatus_CycleMinimumTime" Value="5" />
  <Parameter Name="MachineStatus_CycleExtension" Value="100" />
  <Parameter Name="ConfigMDEJtp_JtpConnector" Value="JtpConnector" />
  <Parameter Name="ConfigMDEJtp_ListServiceName" Value="WorkplaceStatusAssignment.list" />
  <Parameter Name="ConfigMDEJtp_RequestInterval" Value="5" />
</ComponentConfiguration>-->
```

The ***ProductionMonitoringMachineStatusProvider*** component provides the following configuration parameters in addition to the parameters of the parent component ([MachineStatusProvider](#)):

MinDisturbanceInterval

Minimum malfunction time in seconds. After the minimum malfunction time, the machine switches to the status 30000 "not assigned".

Standard *ProductionMonitoringMachineStatusProvider* configuration:

```
<!--<ComponentConfiguration Name="ProductionMonitoringMachineStatusProvider"
      Class="mpdv.MachineDataCollector.Jis.MachineData.ProductionMonitoringMachineStatusProvider"
      Children="OpcDriver">
  <Parameter Name="MachineStatus_MachineStatusConfiguration" Value="wpstatus://HeatingWorkplaceStatusConfiguration.xml" />
  <Parameter Name="MachineStatus_MinDisturbanceInterval" Value="2" />
  <Parameter Name="ConfigMDEJtp_JtpConnector" Value="JtpConnector" />
  <Parameter Name="ConfigMDEJtp_ListServiceName" Value="WorkplaceStatusAssignment.list" />
  <Parameter Name="ConfigMDEJtp_RequestInterval" Value="5" />
</ComponentConfiguration>-->
```

The ***FileTransporter*** component includes the following configuration parameters:

- **InputDirectory:**
Specifies the local directory that stores the files to be transferred (usually identical to the *HydraMDEWriter-OutputDirectory*). If you enter a relative path, the runtime directory is used as the basis.
- **OutputDirectory:**
Specifies a shared directory in the HYDRA server or of a network drive. This directory must be writable. You also have to change the HYDRA configuration in order to read PDV data files from this directory. If you enter a relative path, the runtime directory is used as the basis.

- **SearchPattern:**
Search dialog to restrict the files to be transferred. Use this dialog to define the file extensions that should be transferred by the *FileTransporter*. For *MDE* you can generally use the pattern "*.mdat".
- **Interval:**
Interval in seconds at which DMC checks the input directory for new data files and, if necessary, transfers these files to the output directory.

Standard *MDE FileTransporter* configuration:

```
<ComponentConfiguration Name="HydraMDETransporter"
    Class="mpdv.MachineDataCollector.Core.Components.FileTransporter">
  <Parameter Name="InputDirectory" Value="MDE" />
  <Parameter Name="OutputDirectory" Value="\\server\Hydra1\1\DMC\DMW_01" />
  <Parameter Name="Interval" Value="90" />
  <Parameter Name="SearchPattern" Value="*.mdat" />
</ComponentConfiguration>
```